Discovery of exact equations via computing the Gröbner basis

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In this abstract, we propose a method for addressing the machine learning task of equation discovery (also known as symbolic regression) [12]. We demonstrate its strong connection to a fundamental concept in algebraic geometry: the correspondence between an affine variety and a vanishing ideal. More specifically, we propose that the problem of calculating the Gröbner basis [2] of a vanishing ideal is related to and can be beneficial to the task of equation discovery. While algorithms derived from Buchberger's algorithm have been previously utilized to discern systems of ordinary differential equations [6], we believe that the realm of equation discovery, particularly the segment about algebraic equations (including the so-called exact equations), has not yet been explored by such algorithms. To illustrate the potential of this approach, we focus on an application of equation discovery that aligns more closely with the perspective of a mathematician. This potential is exemplified by recent advancements in applying machine learning techniques in theoretical algebraic geometry [5]. To demonstrate the potential relevance and utility of our approach, we delve into the realm of integer sequences to discover or reconstruct already known exact (recursive) equations, such as the one defined by Fibonacci, using only the given terms of the sequence.

Our first contribution is *MoadeeB*, a novel algorithm for the discovery of exact equations that extends the Möller-Buchberger algorithm. Our second contribution is an extensive evaluation showing *MoadeeB*'s performance and efficiency against state-of-the-art methods while recovering recurrent linear and nonlinear equations from more than 30,000 integer sequences in the Online Encyclopedia of Integer Sequences [11].

Equation discovery, also known as symbolic regression, is a discipline of machine learning focused on the computational discovery of the most fitting and simplest mathematical expression that describes a given data set. Typically, the input data are presented in tabular format as an observation matrix $X \in \mathbb{R}^{m \times n}$, where each row X_i signifies a data point (also referred to as an example within the data set), and each column X^j represents the values of a corresponding variable x_j. The task of discovering *exact* equations from data can be formalized as the objective to find a nontrivial equation:

$$E(x_1, x_2, \ldots, x_n) = 0$$

involving variables x_j that:

- Holds for the data set X exactly;
- Minimizes the complexity of the expression $E(x_1, x_2, \ldots, x_n)$, e.g., number of nodes in its abstract syntax tree.

This task is guided by Occam's razor principle [4], which advocates for the simplest explanation. Symbolic regression aims to automate the scientific discovery process, as exemplified by the manual search for equations that best describe data, practiced by scientists like Kepler, Newton, or Maxwell.

Algebraic geometry [8], on the other side, studies solutions to systems (sets) of polynomial equations, known as algebraic varieties. These varieties and the ideals (sets of polynomials) associated with them in a ring (an algebraic structure extending the concept of integers) can be infinitely large. However, the elegance of algebraic geometry lies in its ability to simplify even such complex structures. Through the concept of ideal basis — a possibly finite set of polynomials — entire (possibly infinite) ideals can be represented, effectively condensing the infinite into a manageable form. This fundamental process of simplification allows mathematicians to delve into the intricate relationships between ideals and varieties, unlocking the rich landscape of algebraic structures.

In the interest of precision, without going into all the mathematical intricacies [8], we first define the *vanishing ideal*, denoted $I(X)$, corresponding to a subset X of the so-called *affine space* $\mathbb{A}^n = K \times \cdots \times K$ over an algebraically closed field, such as $\mathbb{C}$. The vanishing ideal $I(X)$ is defined as the collection of all multivariate polynomials f on $X_1, ..., X_n$, that is, $f \in K[X_1, ..., X_n]$, which equate to zero (i.e., vanish) at every point $(x_1, ..., x_n)$ within the specified subset $X \subset K^n$. In other words,

$$I(X) := \{f \in K[X_1, ..., X_n] \mid f(x_1, ..., x_n) = 0 \text{ for all } (x_1, ..., x_n) \in X\}.$$

It is worth mentioning that the set X is termed as the *affine variety* of the ideal $I(X)$, provided it includes all common zeros of all elements of $I(X)$. The link between equation discovery and algebraic geometry can be elucidated as follows: According to the definition above, for each element f of the ideal $I(X)$, the equation

$$f(x_1, ..., x_n) = 0$$

is valid for all points $x = (x_1, ..., x_n)$ in X. Consequently, if the set X is viewed as the input data set and only polynomial expressions are taken into account, the equation discovery problem can be reinterpreted as the task of identifying the simplest element of the associated vanishing ideal. While the scope of equation discovery is not confined solely to polynomials, the utilization of algebraic geometry tools is still a viable approach, particularly in the absence of any background knowledge or constraints on the data.

The simplification of the vanishing ideal, as previously discussed, involves identifying a finite set of polynomials that not only form a basis for the ideal, but also satisfy specific conditions. A prime example of such a basis is the *Gröbner basis* [2]. This basis is characterized by additional conditions, defined by a so-called monomial order, which essentially necessitates the lowest degrees of the polynomials in the basis. Calculating the Gröbner basis is a well-established problem in commutative algebra, typically resolved using *Buchberger's algorithm* [2]. Given the uniqueness of the Gröbner basis (subject to the choice of a monomial order), it emerges as a natural solution to the equation discovery problem.

To discover exact equations, we employ the Möller-Buchberger algorithm [7], a method closely related to Buchberger's algorithm that computes the Gröbner basis directly from the given data points. The resulting Gröbner basis is effectively a list of polynomials that serve as candidate equations. We then filter and rank these polynomials by complexity, using the number of terms and the magnitudes of their coefficients as criteria. This procedure yields a shortlist of the simplest candidate equations that fit the data exactly.

Next, we highlight the most relevant results—those for the so called *core* OEIS sequences—summarised in Table 1. We largely follow the experimental design from [3] for comparison against the state-of-the-art exact equation discovery approach Diofantos [3] and SINDy [1]. The two complementary experimental series (consisting of 27,236 and 10,000 OEIS sequences, respectively) support the same overall conclusion, although the performance gap is narrower (and somewhat unclear) and *MoadeeB* no longer outpaces the other methods as dramatically. For core OEIS sequences, *MoadeeB* outperforms Diofantos by reconstructing nine additional equations and discovering two previously unknown ones, which were rigorously proved and subsequently reviewed and accepted by the OEIS curators [10,9]. This represents a notable and significant improvement, as Diofantos and SINDy previously exhibited similar performance on this data set, with only a two-equation difference in successful reconstructions using SINDy's default settings. The improvement on the core sequences is significant also because it has the potential to yield discoveries for sequences deemed very important by OEIS curators and, consequently, are the focus of ongoing mathematical research.

We epmhasize the scientific discovery of two novel recurrence relations that resulted from the application of our method. For the sequence of binary tree counts by height (OEIS A001699), we identified the following recurrence:

$$a_n = -a_{n-2}^3 + a_{n-1}^2 + 3a_{n-1}a_{n-2} + 2a_{n-2}^2 + 2a_{n-1} - 4a_{n-2} \ .$$

Similarly, for the sequence of planted 3-tree counts (OEIS A002658), we discovered the relation:

$$a_n = -\frac{a_{n-2}^3 - 2a_{n-1}^2 - 6a_{n-1}a_{n-2} - 4a_{n-2}^2 - 8a_{n-1} + 11a_{n-2}}{4} \ .$$

Table 1. Comparison of *MoadeeB*, Diofantos, and SINDy performance on the core OEIS sequences, grouped by category.

Category	***MoadeeB***	**Diofantos**	**SINDy-default**	**SINDy-tuned**
Trivial $\vee$ Hard $(4 + 94)$	4	4	4	4
Simple (66)	43	32	30	31
All (164)	47	36	34	35

As mentioned above, these two previously unknown equations have since been rigorously proved and formally accepted into the OEIS [10,9].

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